Personal AI That Learns to Work With Its User

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Research ambition

I want to build personal AI that becomes a better collaborator through shared work. Such a partner would understand the current situation, connect it with previous conversations and joint experience, investigate questions, compare alternatives, and carry decisions through to action. My long-term ambition is an assistant that grows familiar with a person's working rhythm and support needs while helping them accomplish increasingly substantial tasks. At AI4SG, I want to advance the personal understanding and sustained collaboration that make this possible.

My first system will be a personal work partner connecting conversation, tasks, calendars, working documents, and software tools. It will help a user investigate a problem, develop a plan, execute agreed actions, and review what happened. Screen activity, speech, and other contextual signals can enrich this shared understanding; spatial and tangible interfaces offer further ways to participate in the user's activity. The research centres on how experience changes the assistant's next decision.

Turning personal understanding into a basis for action

My central question is: how can an assistant transform interaction and task outcomes into revisable understanding of a person, then use that understanding to improve subsequent plans? I will study persistent connections between a specific personal inference, its conditions of applicability, the planning decisions that used it, and their execution outcomes.

A strong comparison system will store searchable interaction and task records, update contextual user entries with uncertainty markers from feedback, and retrieve relevant memories during planning. My proposed system will additionally record which version of a personal inference actually informed each decision. An unexpected outcome or user correction will trigger a focused review of that inference. Context and the user's explanation will guide whether to retain it, narrow its scope, or mark it for clarification. The system will then revisit pending arrangements that still depend on the earlier version.

For example, an assistant may estimate an hour for a familiar writing task. If the task runs longer because it requires additional literature searches, the assistant can distinguish research-intensive writing, revise that estimate, and check upcoming tasks scheduled using the original assumption. The outcome identifies what needs investigation; the task context and user explanation determine the revision. This makes a correction consequential for future work.

Research contribution

PAHF integrates clarification, retrieved preferences, and post-action feedback for continual personalization [1]. ClueUp studies uncertain user profiles [2], while PROUR studies proactive information acquisition [3]. Building on these foundations, I will investigate the persistent links between personal understanding and the decisions it supports, together with targeted revision of dependent plans. My hypothesis is that these links will reduce inappropriate reuse of experience under changing conditions while preserving the value of experience that remains applicable. The contribution will combine an executable collaboration system, an inspectable updating mechanism, and evidence about task assistance and sustained human-AI cooperation.

Implementation and evaluation

Formative interviews and activity observation will identify work tasks in which people repeatedly explain background and where personal experience affects later decisions. I will implement the prototype with an existing language model, structured experience records, and software tools. Three conditions will compare the strong baseline, the complete system, and a version that retains the explicit links while disabling targeted revision. All conditions will retain ordinary feedback-based memory updates.

The mechanism study will replay identical recorded episodes, including user statements, clarification answers, past actions, and their corresponding outcomes. Each system will then receive the same new task and factual evidence. Models, computation, and tool budgets will be matched; retrieval and organisation will follow each method. This stage fixes the source evidence without introducing additional personal information through free-form questioning. Outcomes of new actions will be evaluated against the actions each system actually takes.

The primary measure will be inappropriate reuse of experience after a contextual change: whether an outdated or inapplicable inference changes the resulting plan. Scoring will follow task goals, known constraints, and predefined applicability conditions. Additional measures will assess preservation of valid experience, task completion, and arrangements requiring correction. This comparison directly tests whether the proposed connections improve subsequent decisions under matched information.

An interactive study will assign predefined clarification budgets and randomise system conditions within each budget level. User responses will follow a common scenario fact table. I will compare task outcomes within assigned budgets and report actual questions, facts acquired, correction time, and tool costs together. A subsequent multiweek study will examine real work, including replanning, proactive help, repeated explanations, familiarity, co-presence, and perceived control. Matched replay will support mechanism-level conclusions; interactive and longitudinal studies will establish how the complete system performs in use. Screen and audio inputs will be introduced separately to measure the value of additional contextual evidence. Pilot work and the primary outcomes will guide sample planning.

Preparation and research trajectory

My background combines a five-year architecture degree with a Master of Computer Science expected in December 2026. In Signal Box, I developed the prototype and printed interaction materials and facilitated a course user-testing session. This gave me experience connecting an interaction concept with observation and feedback.

In Game Fellows, I built a working companion and conducted an informal trial with ten participants. Each experienced solo play, a basic Fellow, and a Fellow connected to visual AI. The AI used a stream of game screenshots to judge positions, make action decisions, and provide spoken feedback. Comments about a sense of companionship prompted my interest in how an agent’s participation connects practical assistance with the experience of working together.

In an eight-person image-detection project, I independently reconstructed 900 fusion decisions from frozen model outputs. Analysing a specific disagreement subset revealed asymmetric error costs hidden by the aggregate improvement. Alongside my experience integrating spatial displays and on-site systems in CAVE, this work equips me to combine interactive prototyping, reproducible analysis, and studies of actual use. I want to turn that combination into personal AI that learns how to work with its user.

[1] Liang et al. Learning Personalized Agents from Human Feedback. 2026. https://arxiv.org/abs/2602.16173

[2] ClueUp. ICASSP 2026. https://doi.org/10.1109/ICASSP55912.2026.11464552

[3] Li et al. Clarify the User or Verify the World? 2026. https://arxiv.org/abs/2609.32255